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The Ta-B system: Key experiments and thermodynamic modeling

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ABSTRACT

In the present work, the Ta-B system was thermodynamically optimized based on the CALPHAD method using data of *liquidus* temperatures, invariant transformations and enthalpy of formation from the literature as well as new experiments performed in our group. The phases Ta_3B_2, TaB(rt), Ta_5B_6, Ta_3B_4 were modeled as stoichiometric compounds. On the other hand, L (liquid), BCC, Ta_2B, TaB(ht) and TaB_2 were modeled as solution phases using the Compound Energy Formalism (CEF), with excess terms described by the Redlich-Kister polynomials. (βB) was treated as a pure element without any Ta solubility. The present thermodynamic modeling is in good agreement with the experimental data.

1. Introduction

The development of new high-temperature structural materials is essential for future generations of aircraft engines and land-based gas turbines. RM-boron-silicide-based alloys (RM: refractory metal) represent good candidates for high-temperature applications due to a suitable balance of properties required in these severe conditions [1,2].

The currently accepted Ta-B phase diagram [3], shown in Fig. 1, is mainly based on the proposals of Rudy [4] and Portnoi et al. [5]. However, more recent data from Chad et al. [6,7] have shown that: (i) the liquid composition in the Ta-rich eutectic reaction ($L \leftrightarrow Ta_{ss} + Ta_2B$) is placed at 18 at% B instead of 23 at% B; (ii) the liquid composition in the Ta-rich peritectic reaction ($L + TaB \leftrightarrow Ta_2B$) is placed at 22.5 at% B instead of 27 at% B; (iii) the eutectoid decomposition of Ta_2B occurs at 1925 ± 25 °C instead of 2040 °C; (iv) unknown peaks appears on diffractograms of as-cast Ta-B alloys with composition in the 33–45 at% B range but not in samples with B contents higher than 50 at% B. Furthermore, Okada et al. [8,9] proposed the stability of Ta_5B_6 in the Ta-B system and discrepancies exist about the homogeneity range of TaB_2 [4,8]. In the present work, experiments were designed to investigate these aspects.

The thermodynamic data available for this system concern the experimental determination of TaB_2 enthalpy of formation [10–13] and its dependence on temperature [14]. In addition, there are data on the enthalpies of formation of the intermetallic compounds from ab-initio calculations [15–19].

The first optimization of the Ta-B system available in the literature is due to Kaufman [20]. In his work: (i) all solid phases were modeled as stoichiometric compounds; (ii) the composition of Ta_2B was considered to be at 33.3 at% B and not between 29 at% B and 31 at% B, as suggested by Rudy [4] and Portnoi et al. [5]. Furthermore, most of the modern thermodynamic databases use the Stable Element Reference (SER), adopted by SGTE (Scientific Group Thermodata Europe), which is not compatible with the reference state adopted by Kaufman [20]. In a recent work, Yang et al. [19] (Fig. 2) re-optimized the Ta-B system considering the data of Chad et al. [6] except for the non-stoichiometric composition of Ta_2B. Thus, based on the previous information and the experimental data from the present work, a new thermodynamic modeling for the Ta-B system has been carried out.

2. Experimental procedure

In this work, key experiments were carried out in order to: (i) determine the homogeneity range of TaB_2 phase at 1500 °C; (ii)

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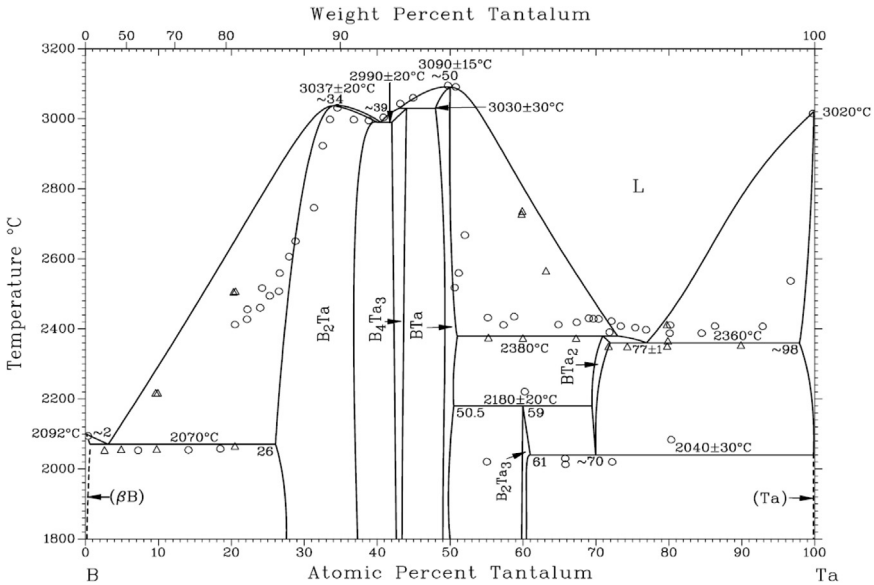

Fig. 1. Currently accepted phase diagram of the Ta-B system [3] along with experimental data from: ○ Rudy [4] and Δ Portnoi et al. [5].

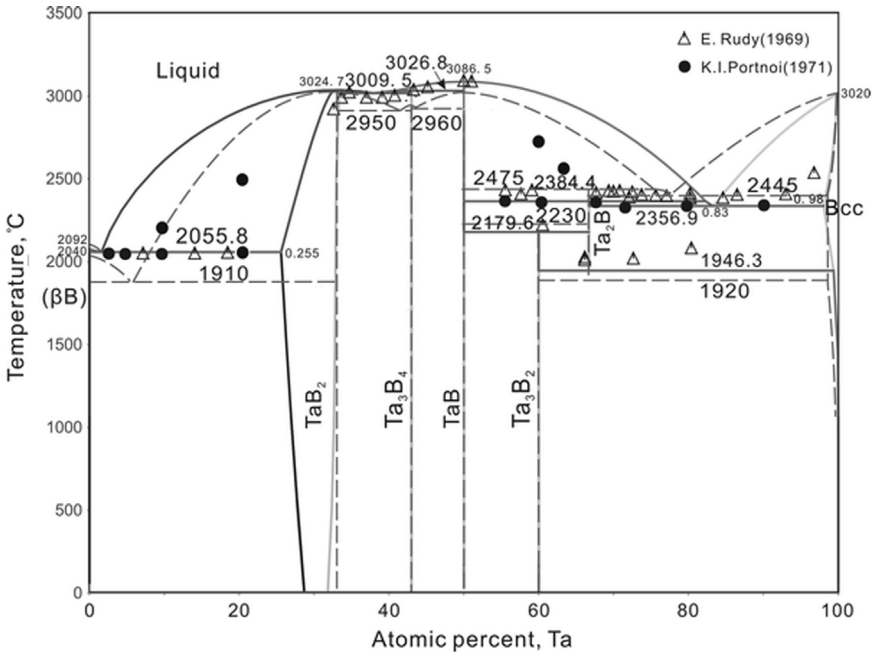

Fig. 2. Ta-B phase diagram published by Yang et al. [19] calculated with their parameters (solid lines) and Kaufman's [20] (dotted lines) along with selected experimental data from Rudy [4] and Portnoi et al. [5].

investigate the possible existence of a high-temperature polymorph TaB-phase; (iii) evaluate the possible stability of Ta_5B_6. Arc-melted samples were used for (i) and (ii) investigations and powder sintered samples for (iii), as detailed below.

Arc-melted samples were prepared from high-purity Ta (min. 99.95 wt%) and B (min. 99.5 wt%) in powder form. The powders were intimately mixed and cold pressed in the form of pellets (~ 3 g) and then arc-melted under titanium-gettered argon (min 99.995 wt%) on a water-cooled copper hearth using a non-consumable tungsten electrode. At least three melting steps were carried out for better chemical homogeneity of the samples.

For the determination of the homogeneity range of TaB_2, arc-melted alloys with compositions varying from 64 to 78 at% B were heat treated under high-purity, niobium-gettered argon, at 1500 °C for 6 h. The solubility limits of TaB_2 were evaluated through the determination of lattice parameters of this phase from the different samples. Within the single-phase region, the lattice parameters of a non-stoichiometric phase varies with composition and depends on the defect structure of the particular phase (substitutional atoms; interstitial atoms; vacancies). When the solubility limit is reached, new phases precipitate (Ta_3B_4 or (βB) in this case) and the lattice parameters of the considered phase (TaB_2) is kept constant in each of the neighbor two-phase fields (Ta_3B_4 + TaB_2; TaB_2 + (βB)).

The investigation of the existence of a high temperature TaB-phase was based on an arc-melted alloy with composition 45 at% B (referred as Ta-45at%B) analyzed in the as-cast state.

For the Ta_5B_6 stability study, two alloys with compositions 54.54 and 60 at% B (referred as Ta-54.54at%B and Ta-60at%B, respectively) were prepared from pure B (min. 99.5 wt%) and TaB powders. The TaB powder was prepared by arc-melting pure Ta and B, following the same

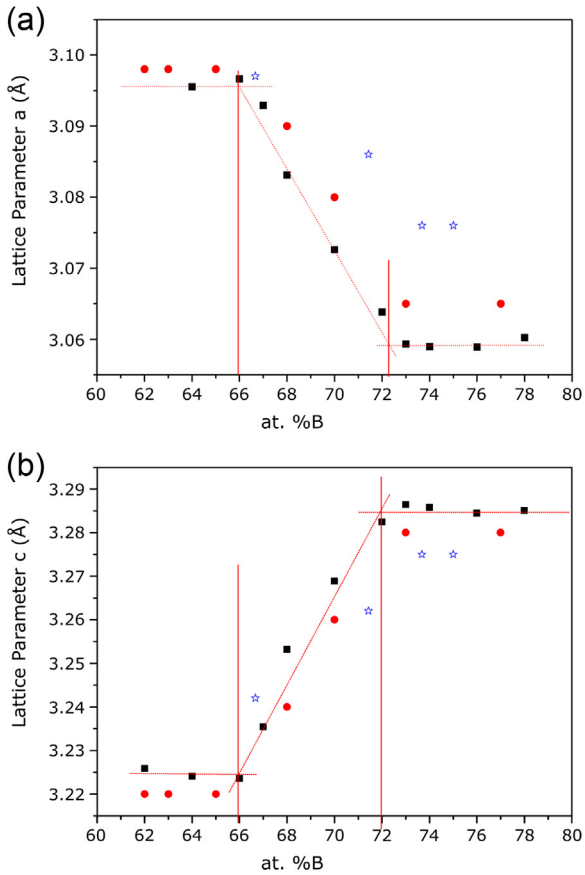

Fig. 3. (a) Variation of the parameter "a" and (b) parameter "c" of the TaB_2 phase in function of B content in the Ta-B system. Symbols are from: ● Rudy [4] at 2000 °C, ☆ Okada [8] at 1650 °C and ■ This work at 1500 °C.

procedure described above, then ground and sieved to below 44 μm. The blended mixture of powders were cold pressed in the form of pellets (~ 3 g) and heat-treated under high-purity, niobium-gettered argon, at 1400 °C/3 h, 1600 °C/2 h and 1800 °C/2 h. The sample with composition Ta-60at%B was prepared to compensate B loss during heat treatment.

All processed alloys were characterized via X-ray powder diffraction. Pieces of the samples were mechanically ground and sieved to below 325 mesh (44 μm). The data were collected with Cu-Kα radiation and the phases identified using data in Villars and Calvert [21] and the Powder Cell software [22].

3. Experimental results

3.1. TaB_2 homogeneity range

Fig. 3 shows the lattice parameters of the TaB_2 phase after heat treatment at 1500 °C along with the experimental data from Rudy [4] (samples heat-treated at 2000 °C) and Okada [8] (samples heat-treated at 1650 °C). As can be observed, the homogeneity range of TaB_2 at 1500 °C varies from 66 to 72 at% B, instead of 62 and 72 at% B [8], in agreement with the values reported by Rudy [4]. This homogeneity range is close to the one proposed for the NbB_2-phase (65–70 at% B) [23].

3.2. High temperature TaB polymorph

Fig. 4 shows the X-ray diffractogram of the as-cast Ta-45at%B alloy. Peaks related to BCC, Ta_2B, TaB(rt) and TaB(ht) (FeB prototype) phases were indexed. The presence of four phases in the microstructure of this sample is related to the solidification in non-equilibrium conditions. The proposition of the stability of TaB(ht) in the Ta-B binary is corroborated by the observation of this FeB-prototype for compositions near TaB in the Ta-Ti-B and Ta-Hf-B ternary systems [24] and the report of unknown peaks in the X-ray diffractograms of several binary as-cast Ta-B alloys in previous works [6,7].

3.3. Ta_5B_6 stability

The XRD results of both Ta-54.54at%B and Ta-60at%B alloys presented peaks from Ta_5B_6, indicating that this phase is stable at temperatures between 1400 and 1800 °C. Fig. 5 shows the X-ray diffractogram of the Ta-60at%B alloy heat-treated at 1600 °C/2 h exhibiting peaks related to TaB(rt), Ta_5B_6, Ta_3B_4 and TaB_2 phases. It is clear that thermodynamic equilibrium conditions have not been reached, however, the presence of Ta_5B_6 supports the claim of its stability in the Ta-B system.

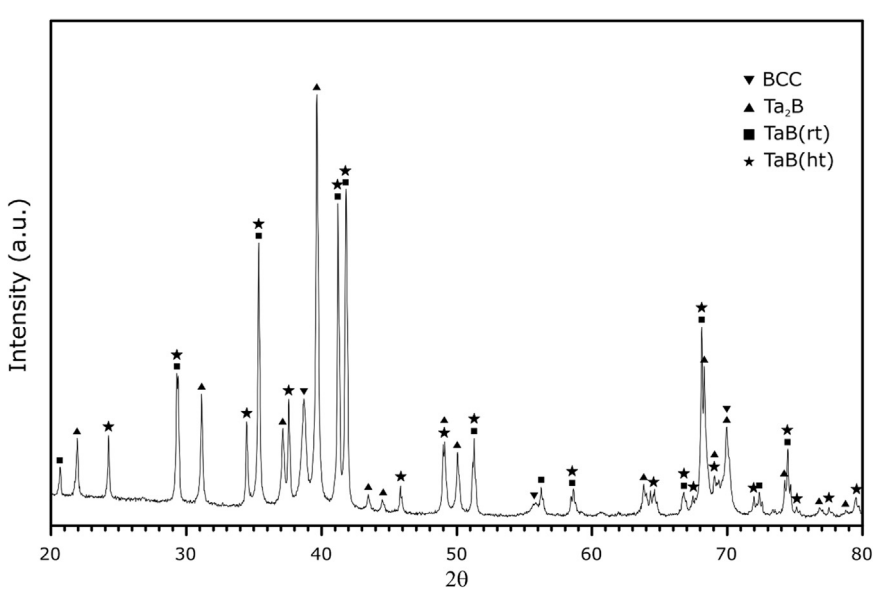

Fig. 4. X-ray diffractogram of the as-cast Ta-45at%B alloy.

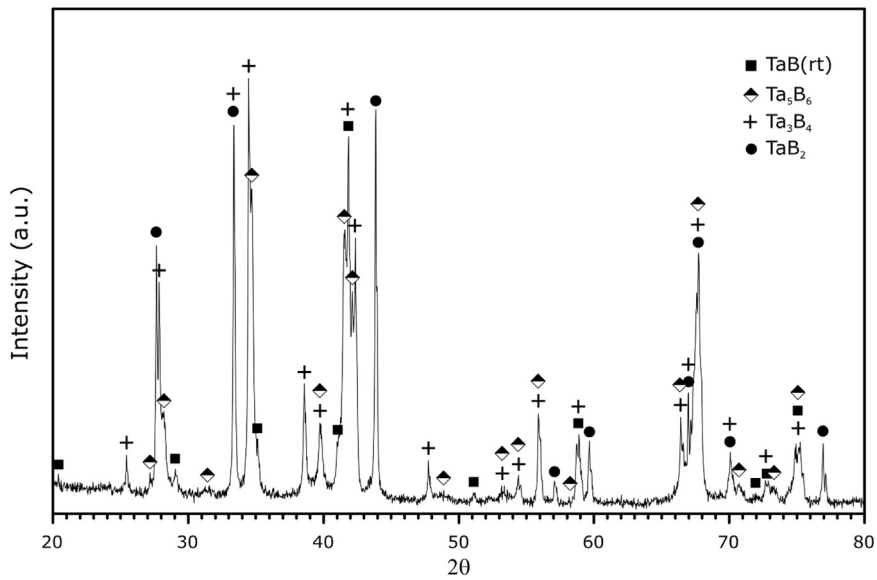

Fig. 5. X-ray diffractogram of the alloy Ta-60at%B heat-treated at 1600 °C/2 h.

4. Thermodynamic modeling

Table 1 shows the crystal structure data of the stable solid phases in the Ta-B system, at their ideal stoichiometry [21]. Employing the Parrot module of the Thermo-Calc software the phases Ta_3B_2, TaB(rt), Ta_5B_6, Ta_3B_4 were modeled as stoichiometric compounds. The phases L (liquid), BCC, Ta_2B, TaB(ht) and TaB_2 were modeled as solutions, using the Compound Energy Formalism (CEF) described in detail by Sundman and Ågren [25] as well as by Hillert [26], with excess terms described by the Redlich-Kister polynomials.

4.1. Gibbs energy for pure components

The temperature dependent Gibbs energy of each pure element is described by an equation of the general form:

$$^{o}G_{m}^{\varphi} - H_{i}^{SER} = a + bT + cT\ln(T) + dT^{2} + eT^{3} + fT^{-1} \quad (1)$$

where H_i^{SER} is the enthalpy of stable element "*i*" at 298.15 K.

4.2. Solution phase liquid

The liquid phase was modeled as a single sublattice substitutional solution. The molar fractions and the end-members are coincident with the pure elements, i.e. $^{o}G_{end1}^{L} = {}^{o}G_{Ta}^{L}$ and $^{o}G_{end2}^{L} = {}^{o}G_{B}^{L}$, yielding the following expression for its Gibbs free energy:

$$G_{m}^{L} = x_{Ta}\,{}^{o}G_{Ta}^{L} + x_{B}\,{}^{o}G_{B}^{L} + RT\left[x_{Ta}\ln(x_{Ta}) + x_{B}\ln(x_{B})\right] + x_{Ta}x_{B}\left[{}^{0}L_{Ta,B}^{L} + {}^{1}L_{Ta,B}^{L}(x_{B}-x_{Ta}) + {}^{2}L_{Ta,B}^{L}(x_{B}-x_{Ta})^{2}\right] \quad (2)$$

Table 1
Crystal structure data of the stable solid phases of the Ta-B system [21].

Phase	prototype	space group	Pearson symbol	Wyckoff position	atom	x	y	z
(Ta), BCC	W	*Im-3m*	*cI*2	2a	Ta	0	0	0
				6b	*ip*[a]	0	0.5	0.5
Ta_2B	Al_2Cu	*I4/mcm*	*tI*12	8h	Ta	0.1661	0.6661	0
				4a	B	0	0	0.25
Ta_3B_2	U_3Si_2	*P4/mbm*	*tP*10	2a	Ta	0	0	0
				4g	B	0.389	0.889	0
				4h	Ta	0.181	0.681	0.5
TaB(rt)	CrB	*Cmcm*	*oC*8	4c	Ta	0	0.1453	0.25
				4c	B	0	0.4360	0.25
TaB(ht)	FeB	*Pnma*	*oP*8	4c	Ta	0.1758	0.25	0.6239
				4c	B	0.0370	0.25	0.111
Ta_5B_6	V_5B_6	*Cmmm*	*oS*22	4j	B	0	0.0832	0.5
				4j	Ta	0	0.1943	0.5
				4j	B	0	0.4603	0.5
				4i	B	0	0.2734	0
				4i	Ta	0	0.3855	0
				2a	Ta	0	0	0
Ta_3B_4	Ta_3B_4	*Immm*	*oI*14	2c	Ta	0.5	0.5	0
				4g	Ta	0	0.180	0
				4g	B	0	0.375	0
				4h	B	0	0.444	0.5
TaB_2	AlB_2	*P6/mmm*	*hP*3	1a	Ta	0	0	0
				2d	B	0.3333	0.6667	0.5
(βB)	B	*R-3m*	*hR*111	–[b]	B	–	–	–

[a] *ip* stays for octahedral interstitial positions.
[b] Wyckoff positions and their coordinates were intentionally suppressed for (βB). The complete list can be found in [21].

4.3. Solution phase BCC

The body centered cubic (BCC) crystal structure may be modeled as a phase with two sublattices, namely the sublattice corresponding to 2a Wyckoff positions (substitutional) and that corresponding to interstitial octahedral Wyckoff positions 6b. For the BCC phase in the Ta-B system, it is well known that B occupies the interstitial sites [27], therefore, the model (Ta)(B,Va)$_3$ was chosen. Besides, this is the model adopted to describe this phase in similar systems [28,29], yielding the following expression for its Gibbs energy:

$$G_{\mathrm{m}}^{\mathrm{BCC}} = y_{\mathrm{Ta}}^{\mathrm{s1}} y_{\mathrm{Va}}^{\mathrm{s2}}\, {}^{o}G_{\mathrm{Ta:Va}}^{\mathrm{BCC}} + y_{\mathrm{Ta}}^{\mathrm{s1}} y_{\mathrm{B}}^{\mathrm{s2}}\, {}^{o}G_{\mathrm{Ta:}B}^{\mathrm{BCC}} + 3RT\left[y_{\mathrm{B}}^{\mathrm{s2}} \ln(y_{\mathrm{B}}^{\mathrm{s2}}) + y_{\mathrm{Va}}^{\mathrm{s2}} \ln(y_{\mathrm{Va}}^{\mathrm{s2}})\right] + +y_{\mathrm{B}}^{\mathrm{s2}} y_{\mathrm{Va}}^{\mathrm{s2}} \left[{}^{0}L_{\mathrm{Ta:}B,\mathrm{Va}}^{\mathrm{BCC}} + {}^{1}L_{\mathrm{Ta:}B,\mathrm{Va}}^{\mathrm{BCC}} (y_{\mathrm{B}}^{\mathrm{s2}} - y_{\mathrm{Va}}^{\mathrm{s2}}) + \ldots\right] \quad (3)$$

where ${}^{o}G_{\mathrm{Ta:Va}}^{\mathrm{BCC}}$ and ${}^{o}G_{\mathrm{Ta:}B}^{\mathrm{BCC}}$ correspond to:

$$\begin{aligned} {}^{o}G_{\mathrm{Ta:Va}}^{\mathrm{BCC}} &= {}^{o}G_{\mathrm{Ta}}^{\mathrm{BCC}} \\ {}^{o}G_{\mathrm{Ta:}B}^{\mathrm{BCC}} &= {}^{o}G_{\mathrm{Ta}}^{\mathrm{BCC}} + 3\,{}^{o}G_{\mathrm{B}}^{\mathrm{Rhom}} + \Delta_{\mathrm{f}}^{o} G_{\mathrm{Ta:}B}^{\mathrm{BCC}} \end{aligned} \quad (4)$$

4.4. Solution phase Ta$_2$B

The Ta$_2$B phase exhibits a tetragonal structure (*tI*12, space group *I*4/*mcm*, prototype Al$_2$Cu) with two sublattices, Ta and B atoms occupying Wyckoff 8 h and 4a positions, respectively. This phase shows a homogeneity range (29–31 at% B), which does not include the ideal stoichiometry 2:1 (33.3 at% B). Considering the absence of information about its defect structure and a homogeneity range displaced towards the Ta-rich side (< 33.3 at% B), it is reasonable to suppose that the defect structure is associated to vacancies at B positions. Therefore, the model (Ta)$_2$(B,Va)$_1$ was adopted for Ta$_2$B in the present work, with its Gibbs energy described by the following equation:

$$G_{\mathrm{m}}^{\mathrm{Ta_2B}} = y_{\mathrm{Ta}}^{\mathrm{s1}} y_{\mathrm{B}}^{\mathrm{s2}}\, {}^{o}G_{\mathrm{Ta:}B}^{\mathrm{Ta_2B}} + y_{\mathrm{Ta}}^{\mathrm{s1}} y_{\mathrm{Va}}^{\mathrm{s2}}\, {}^{o}G_{\mathrm{Ta:Va}}^{\mathrm{Ta_2B}} + RT\left[y_{\mathrm{B}}^{\mathrm{s2}} \ln(y_{\mathrm{B}}^{\mathrm{s2}}) + y_{\mathrm{Va}}^{\mathrm{s2}} \ln(y_{\mathrm{Va}}^{\mathrm{s2}})\right] + {}^{E}G_{\mathrm{Ta:}B,\mathrm{Va}}^{\mathrm{Ta_2B}} \quad (5)$$

where:

$$\begin{aligned} {}^{o}G_{\mathrm{Ta:}B}^{\mathrm{Ta_2B}} &= 2\,{}^{o}G_{\mathrm{Ta}}^{\mathrm{BCC}} + {}^{o}G_{\mathrm{B}}^{\mathrm{Rhom}} + \Delta_{\mathrm{f}}^{o} G_{\mathrm{Ta:}B}^{\mathrm{Ta_2B}} \\ {}^{o}G_{\mathrm{Ta:Va}}^{\mathrm{Ta_2B}} &= 2\,{}^{o}G_{\mathrm{Ta}}^{\mathrm{BCC}} + {}^{o}G_{\mathrm{Va}} + \Delta_{\mathrm{f}}^{o} G_{\mathrm{Ta:Va}}^{\mathrm{Ta_2B}} \end{aligned} \quad (6)$$

And, the excess term is given by:

$${}^{E}G_{\mathrm{Ta:}B,\mathrm{Va}}^{\mathrm{Ta_2B}} = y_{\mathrm{B}}^{\mathrm{s2}} y_{\mathrm{Va}}^{\mathrm{s2}}\, {}^{0}L_{\mathrm{Ta:}B,\mathrm{Va}}^{\mathrm{Ta_2B}} \quad (7)$$

4.5. Solution phase TaB(ht)

The TaB(ht) phase was modeled with two sublattices, but there is no experimental information about its defect structure that causes the homogeneity range to displace towards the Ta-rich side (< 50.0 at% B). It is important to note that no extra peaks were found in samples with B content higher than 50 at% B [6,7].

Thus, the lattice defect structure adopted in the present work, namely vacancies at the B-site, is reasonable considering the important differences in the atomic radius of Ta and B. Therefore, the model (Ta)$_1$(B,Va)$_1$ was adopted for TaB (ht) in the present work with its Gibbs energy described by the following equation:

$$G_{\mathrm{m}}^{\mathrm{TaB(ht)}} = y_{\mathrm{Ta}}^{\mathrm{s1}} y_{\mathrm{B}}^{\mathrm{s2}}\, {}^{o}G_{\mathrm{Ta:}B}^{\mathrm{TaB(ht)}} + y_{\mathrm{Ta}}^{\mathrm{s1}} y_{\mathrm{Va}}^{\mathrm{s2}}\, {}^{o}G_{\mathrm{Ta:Va}}^{\mathrm{TaB(ht)}} + RT\left\{\left[y_{\mathrm{Ta}}^{\mathrm{s2}} \ln(y_{\mathrm{Ta}}^{\mathrm{s2}}) + y_{\mathrm{B}}^{\mathrm{s2}} \ln(y_{\mathrm{B}}^{\mathrm{s2}})\right]\right\} + {}^{E}G_{\mathrm{Ta:Va,}B}^{\mathrm{TaB(ht)}} \quad (8)$$

where:

$$\begin{aligned} {}^{o}G_{\mathrm{Ta:}B}^{\mathrm{TaB(ht)}} &= {}^{o}G_{\mathrm{Ta}}^{\mathrm{BCC}} + {}^{o}G_{\mathrm{B}}^{\mathrm{Rhom}} + \Delta_{\mathrm{f}}^{o} G_{\mathrm{Ta:}B}^{\mathrm{TaB(ht)}} \\ {}^{o}G_{\mathrm{Ta:Va}}^{\mathrm{TaB(ht)}} &= {}^{o}G_{\mathrm{Ta}}^{\mathrm{BCC}} + {}^{o}G_{\mathrm{Va}} + \Delta_{\mathrm{f}}^{o} G_{\mathrm{Ta:Va}}^{\mathrm{TaB(ht)}} \end{aligned} \quad (9)$$

And the excess term is given by:

$${}^{E}G_{\mathrm{Ta:}B,\mathrm{Va}}^{\mathrm{TaB(ht)}} = +y_{\mathrm{B}}^{\mathrm{s2}} y_{\mathrm{Va}}^{\mathrm{s2}}\, {}^{0}L_{\mathrm{Ta:Va,}B}^{\mathrm{TaB(ht)}} \quad (10)$$

4.6. Solution phase TaB$_2$

The TaB$_2$ phase exhibits a hexagonal structure (*hP*3, space group *P6/mmm*, prototype AlB$_2$) with two sublattices, Ta and B atoms occupying Wyckoff 1a and 2d positions, respectively. As mentioned before, this phase shows an extensive homogeneity range (66–72 at% B). Post et al. [30] compared the structural characteristics of several diborides and proposed vacancies in both Ta and B positions as defect mechanisms to explain the homogeneity range. This lattice defect structure has been experimentally confirmed as the reason for the homogeneity range of the NbB$_2$ phase in the Nb-B system [23], as well as for Zr-stabilized, nonstoichiometric molybdenum diboride [31]. This lattice defect structure has been successfully used in the thermodynamic modeling of the NbB$_2$ phase [29]. Therefore, the model (Ta,Va)$_1$(B,Va)$_2$ was adopted for TaB$_2$ in the present work with its Gibbs energy described by the following equation:

$$G_{\mathrm{m}}^{\mathrm{TaB_2}} = y_{\mathrm{Ta}}^{\mathrm{s1}} y_{\mathrm{B}}^{\mathrm{s2}}\, {}^{o}G_{\mathrm{Ta:}B}^{\mathrm{TaB_2}} + y_{\mathrm{Ta}}^{\mathrm{s1}} y_{\mathrm{Va}}^{\mathrm{s2}}\, {}^{o}G_{\mathrm{Ta:Va}}^{\mathrm{TaB_2}} + y_{\mathrm{Va}}^{\mathrm{s1}} y_{\mathrm{B}}^{\mathrm{s2}}\, {}^{o}G_{\mathrm{Va:}B}^{\mathrm{TaB_2}} + y_{\mathrm{Va}}^{\mathrm{s1}} y_{\mathrm{Va}}^{\mathrm{s2}}\, {}^{o}G_{\mathrm{Va:Va}}^{\mathrm{TaB_2}} + +RT\left\{\left[y_{\mathrm{Ta}}^{\mathrm{s1}} \ln(y_{\mathrm{Ta}}^{\mathrm{s1}}) + y_{\mathrm{Va}}^{\mathrm{s1}} \ln(y_{\mathrm{Va}}^{\mathrm{s1}})\right] + 2\left[y_{\mathrm{B}}^{\mathrm{s2}} \ln(y_{\mathrm{B}}^{\mathrm{s2}}) + y_{\mathrm{Va}}^{\mathrm{s2}} \ln(y_{\mathrm{Va}}^{\mathrm{s2}})\right]\right\} + {}^{E}G_{\mathrm{Ta,Va:}B,\mathrm{Va}}^{\mathrm{TaB_2}} \quad (11)$$

where:

$$\begin{aligned} {}^{o}G_{\mathrm{Ta:}B}^{\mathrm{TaB_2}} &= {}^{o}G_{\mathrm{Ta}}^{\mathrm{BCC}} + 2\,{}^{o}G_{\mathrm{B}}^{\mathrm{Rhom}} + \Delta_{\mathrm{f}}^{o} G_{\mathrm{Ta:}B}^{\mathrm{TaB_2}} \\ {}^{o}G_{\mathrm{Ta:Va}}^{\mathrm{TaB_2}} &= {}^{o}G_{\mathrm{Ta}}^{\mathrm{BCC}} + 2\,{}^{o}G_{\mathrm{Va}} + \Delta_{\mathrm{f}}^{o} G_{\mathrm{Ta:Va}}^{\mathrm{TaB_2}} \\ {}^{o}G_{\mathrm{Va:}B}^{\mathrm{TaB_2}} &= {}^{o}G_{\mathrm{Va}} + 2\,{}^{o}G_{\mathrm{B}}^{\mathrm{Rhom}} + \Delta_{\mathrm{f}}^{o} G_{\mathrm{Va:}B}^{\mathrm{TaB_2}} \\ {}^{o}G_{\mathrm{Va:Va}}^{\mathrm{TaB_2}} &= {}^{o}G_{\mathrm{Va}} + 2\,{}^{o}G_{\mathrm{Va}} + \Delta_{\mathrm{f}}^{o} G_{\mathrm{Va:Va}}^{\mathrm{TaB_2}} \end{aligned} \quad (12)$$

And the excess term is given by:

$${}^{E}G_{\mathrm{Ta,Va:}B,\mathrm{Va}}^{\mathrm{TaB_2}} = y_{\mathrm{Ta}}^{\mathrm{s1}} y_{\mathrm{Va}}^{\mathrm{s1}} y_{\mathrm{B}}^{\mathrm{s2}}\, {}^{0}L_{\mathrm{Ta,Va:}B}^{\mathrm{TaB_2}} + y_{\mathrm{Ta}}^{\mathrm{s1}} y_{\mathrm{Va}}^{\mathrm{s1}} y_{\mathrm{Va}}^{\mathrm{s2}}\, {}^{0}L_{\mathrm{Ta,Va:Va}}^{\mathrm{TaB_2}} + +y_{\mathrm{Ta}}^{\mathrm{s1}} y_{\mathrm{B}}^{\mathrm{s2}} y_{\mathrm{Va}}^{\mathrm{s2}}\, {}^{0}L_{\mathrm{Ta:}B,\mathrm{Va}}^{\mathrm{TaB_2}} + y_{\mathrm{Va}}^{\mathrm{s1}} y_{\mathrm{B}}^{\mathrm{s2}} y_{\mathrm{Va}}^{\mathrm{s2}}\, {}^{0}L_{\mathrm{Va:}B,\mathrm{Va}}^{\mathrm{TaB_2}} \quad (13)$$

4.7. Stoichiometric phases

The crystal structures of the stoichiometric borides suggest the use of multiple sublattices for their modeling, as follows: (i) for Ta$_3$B$_2$, (Ta)$_2$(B)$_4$(Ta)$_4$; (ii) for TaB, (Ta)$_4$(B)$_4$; (iii) for Ta$_5$B$_6$, (Ta)$_2$(Ta)$_4$(B)$_4$(B)$_4$(Ta)$_4$(B)$_4$; (iv) for Ta$_3$B$_4$, (Ta)$_2$(B)$_4$(Ta)$_4$(B)$_4$. However, in the present work, positions occupied by the same element were considered to be energetically equivalent, reducing their description to 2-sublattices, using the smallest integers as stoichiometric indices. Thus, the Gibbs energies of stoichiometric borides are described by the following equations:

Ta$_3$B$_2$

$$G_{\mathrm{m}}^{\mathrm{Ta_3B_2}} = 3\,{}^{o}G_{\mathrm{Ta}}^{\mathrm{BCC}} + 2\,{}^{o}G_{\mathrm{B}}^{\mathrm{Rhom}} + \Delta_{\mathrm{f}}^{o} G_{\mathrm{Ta:}B}^{\mathrm{Ta_3B_2}} \quad (14)$$

TaB

$$G_{\mathrm{m}}^{\mathrm{TaB}} = {}^{o}G_{\mathrm{Ta}}^{\mathrm{BCC}} + {}^{o}G_{\mathrm{B}}^{\mathrm{Rhom}} + \Delta_{\mathrm{f}}^{o} G_{\mathrm{Ta:}B}^{\mathrm{TaB}} \quad (15)$$

Ta$_5$B$_6$

$$G_{\mathrm{m}}^{\mathrm{Ta_5B_6}} = 5\,{}^{o}G_{\mathrm{Ta}}^{\mathrm{BCC}} + 6\,{}^{o}G_{\mathrm{B}}^{\mathrm{Rhom}} + \Delta_{\mathrm{f}}^{o} G_{\mathrm{Ta:}B}^{\mathrm{Ta_5B_6}} \quad (16)$$

Ta$_3$B$_4$

$$G_{\mathrm{m}}^{\mathrm{Ta_3B_4}} = 3\,{}^{o}G_{\mathrm{Ta}}^{\mathrm{BCC}} + 4\,{}^{o}G_{\mathrm{B}}^{\mathrm{Rhom}} + \Delta_{\mathrm{f}}^{o} G_{\mathrm{Ta:}B}^{\mathrm{Ta_3B_4}} \quad (17)$$

5. Thermodynamic optimization results

The ${}^{n}L_{ij}^{\varphi}$ parameters in the previous equations (${}^{n}L_{ij}^{\varphi}$ = L, BCC, Ta$_2$B, TaB(ht), TaB$_2$) as well as the Gibbs energy of formation of the end-members in Eqs. (4), (6), (9) and (12) and of the stoichiometric

Table 2
Optimized Gibbs energy coefficients for all phases of the Ta-B system. The energy values are given per mole of phase.

Phase	Model	Parameter	Coefficients	
			a	*b*
L	(Ta,B)	${}^{0}L^{L}_{Ta,B}$	− 229,500	+ 52.2
BCC	$(Ta)_1(B,Va)_3$	${}^{0}_{f}G^{BCC}_{Ta:B}$	+ 20,000	
		${}^{0}L^{BCC}_{Ta:B,Va}$	− 12,200	− 3.0
Ta_2B	$(Ta)_2(B,Va)_1$	${}^{0}_{f}G^{Ta_2B}_{Ta:B}$	− 164,800	+ 24.2
		${}^{0}_{f}G^{Ta_2B}_{Ta:Va}$	+ 10,000	
		${}^{0}L^{Ta_2B}_{Ta:B,Va}$	+ 35,900	
Ta_3B_2	$(Ta)_3(B)_2$	${}^{0}_{f}G^{Ta_3B_2}_{Ta:B}$	− 357,000	+ 59.1
TaB(rt)	$(Ta)_1(B)_1$	${}^{0}_{f}G^{TaB(rt)}_{Ta:B}$	− 153,000	+ 19.7
TaB(ht)	$(Ta)_1(B,Va)_1$	${}^{0}_{f}G^{TaB(ht)}_{Ta:B}$	− 136,400	+ 14.6
		${}^{0}_{f}G^{TaB(ht)}_{Ta:Va}$	+ 64,300	
Ta_5B_6	$(Ta)_5(B)_6$	${}^{0}_{f}G^{Ta_5B_6}_{Ta:B}$	− 857,500	+ 113.2
Ta_3B_4	$(Ta)_3(B)_4$	${}^{0}_{f}G^{Ta_3B_4}_{Ta:B}$	− 542,100	+ 70.5
TaB_2	$(Ta,Va)_1(B,Va)_2$	${}^{0}_{f}G^{TaB_2}_{Ta:B}$	− 223,500	+ 28.4
		${}^{0}_{f}G^{TaB_2}_{Ta:Va}$	+ 12,000	+ 2.4
		${}^{0}_{f}G^{TaB_2}_{Va:B}$	+ 100,416	− 19.4
		${}^{0}_{f}G^{TaB_2}_{Va:Va}$	0	
		${}^{0}L^{TaB_2}_{Ta,Va:B}$	− 109,390	+ 25.0
		${}^{0}L^{TaB_2}_{Ta:B,Va}$	+ 247,780	
		${}^{0}L^{TaB_2}_{Ta,Va:Va}$	+ 200,000	
		${}^{0}L^{TaB_2}_{Va:B,Va}$	+ 200,000	

compounds in Eqs. (14)–(17) are described according to Eq. (1). Their coefficients *a* and *b* correspond to adjustable variables in the optimization procedure (see Table 2). The temperature dependent Gibbs energies for the pure elements in their stable and metastable states were taken from SGTE database [32]. Considering the Gibbs energies of formation of the stoichiometric compounds, only the coefficients *a* and *b* of Eq. (1) were used. This is consistent with Neumann-Kopp's rule, which describes the heat capacity of a phase as the sum of the heat capacities of its components, proportional to the phase stoichiometry.

Fig. 6 compares the measurements of Bolgar et al. [14] for the enthalpy, [H(T) – H(298 K)] of TaB_2 (symbols) with values calculated by the Neumann-Kopp's rule (continuous line), indicating that this rule well describes the enthalpy data of this boride.

The current optimization process was mainly based on data of congruent and invariant transformation assessed by Okamoto [3] and the previous data given in item 1 (Introduction) of this paper. Furthermore, the temperatures for the invariant reactions reported by Rudy [4] and Portnoi et al. [5] were also accepted in the present optimization, except for the eutectoid decomposition of Ta_2B which was taken from Chad et al. [6]. The experimental values for the TaB_2 enthalpy of formation reported by Kirpichev et al. [10] and Meschel and Kleppa [11] are −186.575 kJ/mole of phase and −159.900 kJ/mole of phase, respectively. From experiments of evaporation of TaB_2, Leitnaker et al. [12] have estimated that the heat of formation of this boride should be less negative than −190.340 kJ/mole of phase, in agreement with the values measured by Kirpichev et al. [10] and Meschel and Kleppa [11]. The optimization process was performed considering the average of the measured values ([10] and [11]: −173.237 kJ/mole of phase) for the enthalpy of formation of TaB_2.

The model $(Ta)_1(B,Va)_3$ chosen for the BCC leads to the definition of the end-member TaB_3, with its Gibbs energy of formation given by $\Delta_f^o G^{BCC}_{Ta:B}$ as shown in Eq. (4). No experimental information is available for this metastable compound, as a positive value of 5000 J/mole of atoms has been often used, e.g. Ref. [33] for the Gibbs energy of formation of such metastable compounds, the value 20000 J/mole of phase was adopted.

The model $(Ta)_2(B,Va)_1$, chosen for the description of Ta_2B, leads to two end-members given by Eq. (6). The coefficients *a* and *b* for the end-member $\Delta_f^o G^{Ta_2B}_{Ta:B}$ were adjustable variables in the present work. No experimental information is available for the $\Delta_f^o G^{Ta_2B}_{Ta:Va}$ metastable compound, and then the value 10000 J/mole of phase was adopted. The

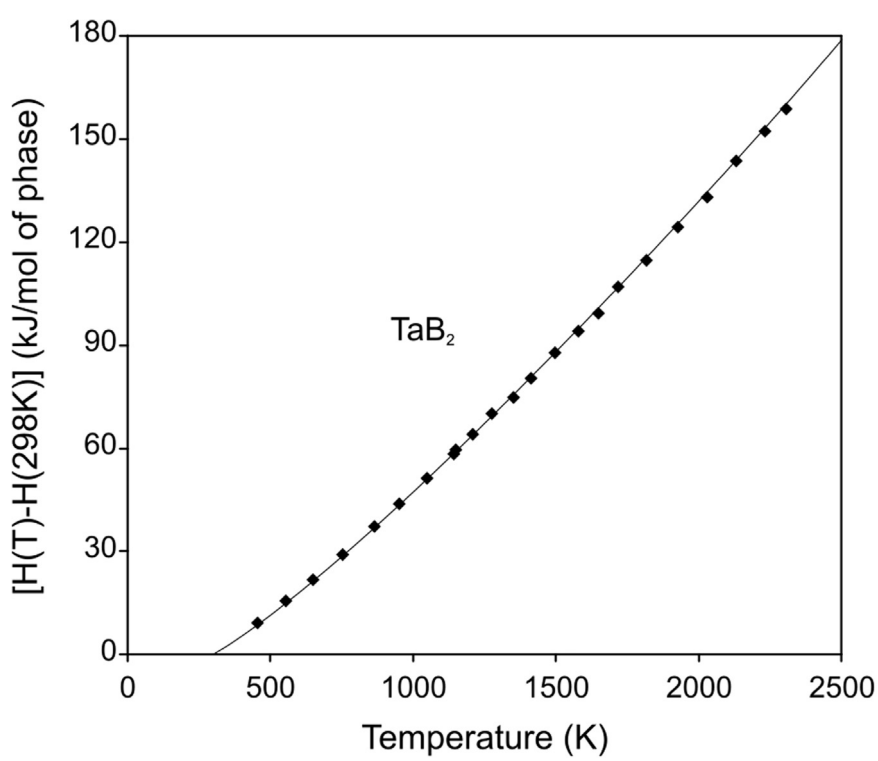

Fig. 6. Calculated (solid line) and experimental (symbols [14]) values of enthalpy variation with temperature [H(T)-H(298)] for TaB_2 given in kJ/mole of phase.

Table 3
Invariant reactions in the Ta-B system.

Invariant Reaction (ϕ_1, ϕ_2, ϕ_3)	Temperature (K)	Composition of B in phases (at%)			Reference
		ϕ_1	ϕ_2	ϕ_3	
L ↔ Ta_2B + BCC	2638	18.7	27.9	2.6	This Work
	2630	17.1	33.3	3.3	Zhang et al. [18]
	–	18	–	–	Chad [6]
	2633	23	28[a]	2	Okamoto [3]
	2633	–	–	–	Portnoi et al. [5]
	2658	23	28[a]	2	Rudy [4]
L + TaB(rt) ↔ Ta_2B	2691	21.2	50.0	29.5	This Work
	2657	17.9	50.0	33.3	Zhang et al. [18]
	–	22.5	–	–	Chad [6]
	2653	27[a]	49[a]	29[a]	Okamoto [3]
	2653	–	–	–	Portnoi et al. [5]
	2690	27[a]	49[a]	30[a]	Rudy [4]
Ta_2B ↔ BCC + Ta_3B_2	2206	30.3	0.6	40.0	This Work
	2219	33.3	0.8	40.0	Zhang et al. [18]
	2198	–	–	–	Chad [6]
	2313	30	0[a]	39	Okamoto [3]
	2313	–	–	–	Portnoi et al. [5]
	2313	30	1[a]	38[a]	Rudy [4]
TaB(rt) + Ta_2B ↔ Ta_3B_2	2445	50.0	30.6	40.0	This Work
	2453	50.0	33.3	40.0	Zhang et al. [18]
	2453	49.5	31[a]	41	Okamoto [3]
	2243[c]	–	–	–	Portnoi et al. [5]
	2453	50[a]	32[a]	40[a]	Rudy [4]
TaB(ht) ↔ TaB(rt) + L	3008	48.7	50.0	28.5	This Work
Ta_3B_4 + TaB(ht) ↔ TaB(rt)	3168	57.1	49.6	50.0	This Work
Ta_3B_4 + TaB(rt) ↔ Ta_5B_6	2848	57.1	50.0	54.5	This Work
L + TaB(ht) ↔ Ta_3B_4	3284	59.5	49.5	57.1	This Work
	3300	59.3	50.0	56.3	Zhang et al. [18]
	3303	57[a]	52[a,b]	55[a]	Okamoto [3]
	3303	59[a]	52[a,b]	56[a]	Rudy [4]
L ↔ TaB_2 + Ta_3B_4	3276	61.6	67.3	57.1	This Work
	3282	62.8	66.6	56.3	Zhang et al. [18]
	3263	61	61[a]	58[a]	Okamoto [3]
	3263	61	63[a]	59[a]	Rudy [4]
L ↔ (βB) + TaB_2	2325	97.8	100.0	72.8	This Work
	2329	98.2	100.0	74.4	Zhang et al. [18]
	2343	98	99[a]	74	Okamoto [3]
	2343	–	–	–	Portnoi et al. [5]
	2328	98	100	74	Rudy [4]

[a] Approximate value obtained by reading the phase diagram.
[b] TaB (rt).
[c] BCC + TaB(rt) ↔ Ta_3B_2.

coefficients for the excess parameter ${}^0L^{Ta_2B}_{Ta:B,Va}$ were adjustable variables in the present work.

The model $(Ta)_1(B,Va)_1$, chosen for the description of TaB(ht), leads to two end-members given by Eq. (9). The coefficients a and b for the end-members $\Delta_f^o G^{TaB}_{Ta:B}$ and $\Delta_f^o G^{TaB}_{Ta:Va}$ were adjustable variables in the present work. The metatectic reaction involving TaB(ht), L and TaB(rt) was modeled in order to obtain a L composition between 28 and 30 at% B. This is the composition in which the extra peaks started to shows up in the samples of Chad et al. [7]. The liquid composition in the metatetic reaction at 29 at% B was chosen as experimental information and the final transformation temperature (3008 K, see Table 3) was consequently adjusted during the optimization procedure. The invariant transformation close to the ideal stoichiometry TaB is supposed to be of peritectoid nature and its temperature was arbitrarily set as experimental information at 3138 ± 20 K. This is a temperature high enough to justify the complete polymorphic transformation TaB(ht) ↔ TaB(rt) and, at the same time, is below the peritectic formation of Ta_3B_4, and above that of the metatectic reaction. The Gibbs energy of TaB(ht) is determined in the present work by the ideal solution of its two end-members.

The proposition of a pericetoid formation for Ta_5B_6 is based on the absence of this phase in as-cast samples [7] as well as its presence in the heat treated samples of the present work. The temperature of this peritectoid reaction has been taken as 2857 K, the mean value between the metatectic and peritectic reactions involving the TaB(rt) phase.

The model $(Ta,Va)_1(B,Va)_2$, chosen for the description of TaB_2, leads to the four end-members given by Eq. (12). The Gibbs energy of the end-member with both sublattices totally vacant was assumed to be 0 ($\Delta_f^o G^{TaB_2}_{Va:Va} = 0$). The SGTE descriptions for hcp-Ta and hcp-B were adopted for the parameters $\Delta_f^o G^{TaB_2}_{Ta:Va}$ and $\Delta_f^o G^{TaB_2}_{Va:B}$, respectively. The coefficients a and b for the end-member $\Delta_f^o G^{TaB_2}_{Ta:B}$ were adjustable variables in the present work. The Gibbs energy of TaB_2 is mainly determined by the reference surface defined by its four end-members (12). During optimization, we achieved the stabilization of this compound at the stoichiometric composition TaB_2 and its vicinity, which is desirable, but also near to compositions correspondent to an almost empty TaB_2 crystal structure with Ta:B ratio of approximately 1:2. To solve this problem, we have arbitrarily attributed a fixed value of 200 kJ/mole of phase for each of the ${}^0L^{TaB_2}_{Ta,Va:Va}$ and ${}^0L^{TaB_2}_{Va:B,Va}$ excess parameters. The coefficients for the other two excess parameters, ${}^0L^{TaB_2}_{Ta,Va:B}$ and ${}^0L^{TaB_2}_{Ta:B,Va}$, were adjustable variables in the present work.

The calculated phase diagram data shown in Fig. 7 and Table 3 consolidates the temperatures and compositions of the phases in the invariant reactions. A good agreement between experimental and calculated values can be observed: (i) The composition of the liquid in the Ta-rich eutectic reported by Chad et al. [6] at 18 at% B was calculated

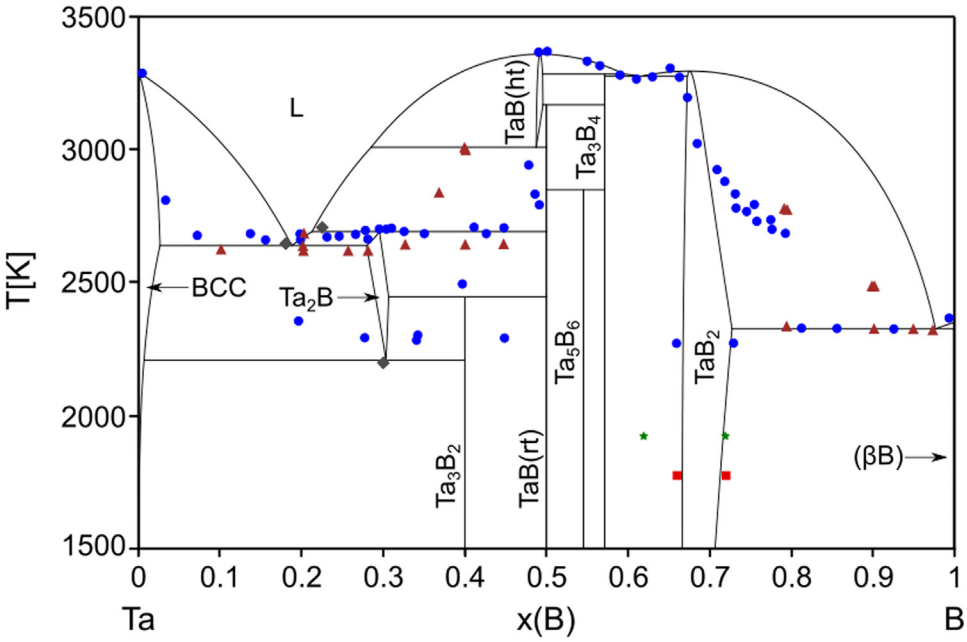

Fig. 7. Phase diagram of the Ta-B system calculated with coefficients from this work along with experimental data from: • Rudy [4], ▲ Portnoi et al. [5], ♦ Chad [6,7], ★ Okada [8] and ■ this work.

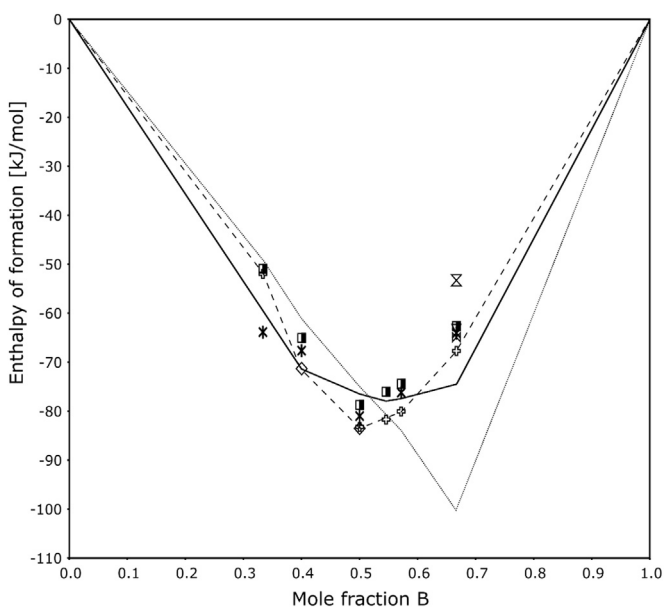

Fig. 8. Enthalpy of formation at 298.15 K in the Ta-B system calculated with the parameters optimized in this work (solid line), by Yang et al. [19] (dashed line) and Kaufman [20] (dotted line) along with experimental data from: ⧖ Meschel and Kleppa [11], ▽ Johnson et al. [13] and ⋈ Kirpichev et al. [10] and ab initio data from: ◨ Kirklin et al. [15] ◇ Qi [16,17], ✚ Zhang et al. [18] and ✱ Yang et al. [19].

at 18.7 at% B; (ii) the composition of the liquid in the Ta-rich peritectic reported by Chad et al. [6] at 22.5 at% B was calculated at 21.2 at% B; (iii) the eutectoid decomposition of Ta_2B reported by Chad et al. [6] at 1925 ± 25 °C (2198 K) was calculated at 1933 °C (2206 K); (iv) the solubility range for TaB_2 at 1500 °C (1773 K) was calculated between 66.67 and 70.7 at% B, near the experimental results (66–72 at% B). Note that our results for the solubility range for TaB_2 at 1500 °C (1773 K) is in agreement with that reported by Rudy at 1650 °C (1923 K) [4].

Fig. 8 shows the enthalpy of formation of the intermetallic phases at 298.15 K. The continuous line represents the values calculated with the parameters optimized in the present work while the dashed line and dotted line are the enthalpies reported by Yang et al. [19] and Kaufman [20] respectively. An important scatter is observed among previous data (experimental as well as DFT) and, therefore, all values for the enthalpy were considered as input information with the same weights in the optimization procedure. The present calculated values agree well with the previous data reported in the literature, except those reported by Kaufman in his thermodynamic modeling.

6. Conclusions

The aim of the present work was to provide a new set of thermodynamic coefficients for the Ta-B system to better reproduce the currently known experimental phase equilibria and thermodynamic data. The phases Ta_3B_2, TaB(rt), Ta_5B_6 and Ta_3B_4 were modeled as stoichiometric compounds. The phases L (liquid), BCC, Ta_2B, TaB(ht) and TaB_2 were modeled as solutions. (βB) was treated as a pure element without any Ta solubility. The calculated phase diagram reproduces well recent experimental information.

Acknowledgements

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Data availability

The datasets analyzed during the current study will be made available from the corresponding author on request.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.calphad.2018.08.012.

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